

# Multiple Hypothesis Testing

*DDA3600: Factor Investing*

**Chuan Shi**
School of Data Science

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香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



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**Multiple hypothesis testing refers to the process of evaluating more than one statistical hypothesis at the same time.**

- In typical hypothesis testing, you assess whether one hypothesis is true or not based on your data.
- When you test multiple hypotheses simultaneously, the chances of making incorrect conclusions (false positives) increase. False positive appears simply due to luck (noise).

**Marcos Lopez de Prado: If the researcher tries a large enough number of strategy configurations, a backtest can always be fit to any desired performance for a fixed sample length.**


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# Intuitive Explanation

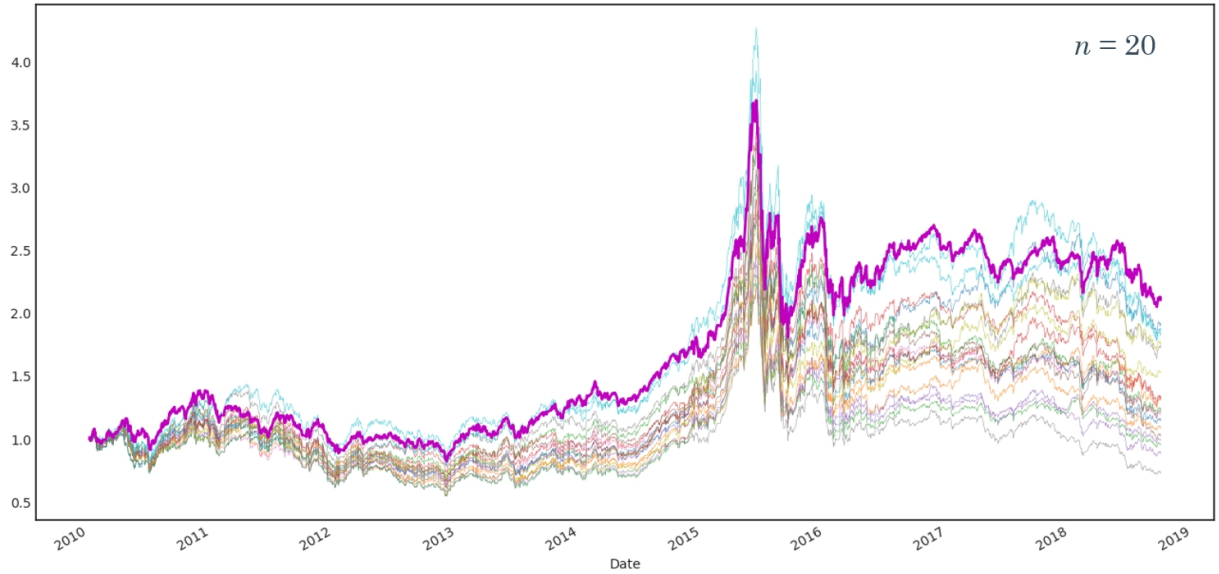


- $p$-value: The *conditional* probability of observing the test statistic or a more extreme value, given that the null hypothesis is true. (For factors, $\mathcal{H}_0 : \mu = 0$.)
- Using sample data of factor returns, construct the $t$-statistic:

$$t = \frac{\bar{x}}{s/\sqrt{T}},$$

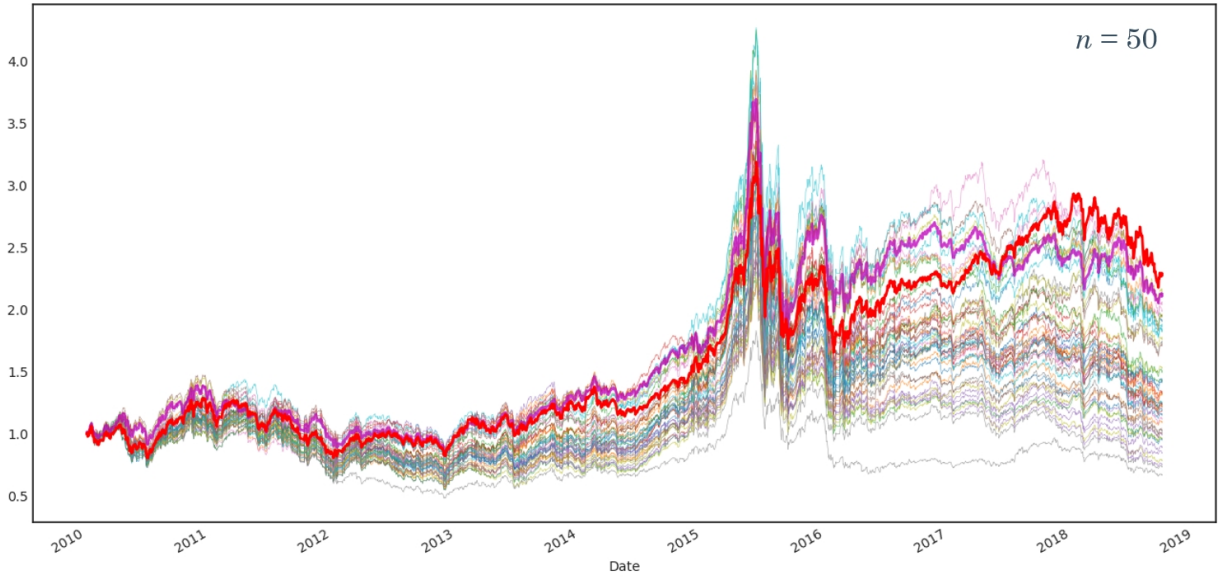
  where $\bar{x}$ is the time series mean of factor returns, $s$ is the sample standard deviation, and $T$ is the number of periods. Then, $p$-value $= \text{prob}(|t_\nu| \geq t_{\text{cutoff}} | \mathcal{H}_0)$.
- In single hypothesis testing, a $p$-value of 0.05 is often used, corresponding to a $|t$-statistic$|$ of approximately 2.0. Suppose that $\mathcal{H}_0$ is true, then the chance that we make a mistake is 5%.
- But consider testing 100 factors simutaneously, then the probability of finding at least one false discovery is as high as $1 - 0.95^{100} > 99\%$.


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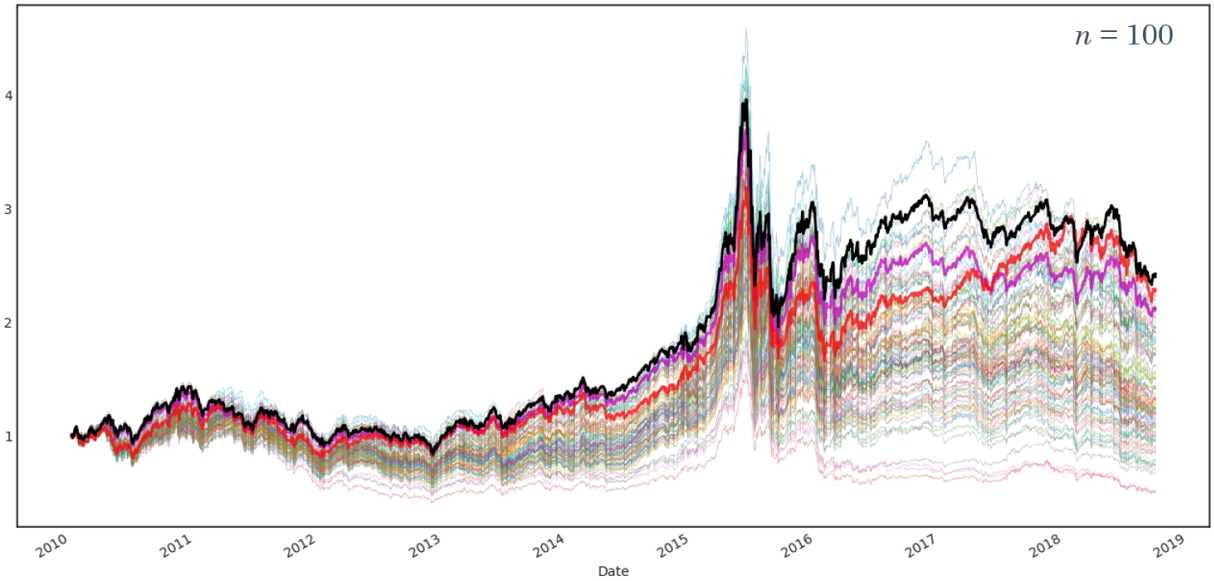


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3.分域建模：传统多因子模型通常进行全市场统一排序，虽可通过行业、市值中性化等方法规避部分差异，却忽略了不同“域”（如宽基、行业、风格、生命周期）内股票特性、参与者行为、驱动因素及定价逻辑的本质区别。分域建模正是基本面量化走向“精细化”的核心体现——在不同领域（如行业、风格、主题）运用差异化因子或模型，提升因子有效性与模型适应性。许多机构会用子策略针对不同市场环境和投资目标进行分域建模

- Mining factors within a specific pool is much easier than doing so across the entire cross-section.
- Extended Thought: Is it reasonable to “mindlessly” mine factors within each individual industry? (Business logic is needed as guidance.)
- Similarly, is it reasonable to divide the sample over time to find the window where factor performance is most significant? This is another form of multiple hypothesis testing. (Let’s look at an example.)


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# An Example





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# An Example, Cont'd

| 样本 | | t-statistic | p-value |
|---|---|---|---|
| All | | 1.1737 | 0.2405 |
| Turn of Month | | -0.1048 | 0.9166 |
| Calendar Month | Jan. | -0.5879 | 0.5566 |
| | Feb. | 1.3948 | 0.1631 |
| | Mar. | **3.0402** | 0.0024 |
| | Apr. | -0.7167 | 0.4735 |
| | May | **2.1774** | 0.0294 |
| | Jun. | 0.1847 | 0.8535 |
| | Jul. | -0.8418 | 0.3999 |
| | Aug. | 0.4468 | 0.6550 |
| | Sep. | 0.5363 | 0.5917 |
| | Oct. | 0.1934 | 0.8466 |
| | Nov. | **2.2958** | 0.0217 |
| | Dec. | **-2.8101** | 0.0050 |
| Day of Week | Monday | -1.0205 | 0.3075 |
| | Tuesday | **2.1189** | 0.0341 |
| | Wednesday | 1.1713 | 0.2415 |
| | Thursday | 0.0664 | 0.9471 |
| | Friday | 0.6660 | 0.5054 |


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# An Example, Cont'd

| 样本 | | 原始 p-value | BHY 调整后 p-value | |
|---|---|---|---|---|
| Turn of Month | | 0.9166 | | 0.9471 |
| Calendar Month | Jan. | 0.5566 | 0.8535 | 0.9471 |
| | Feb. | 0.1631 | 0.8535 | 0.9471 |
| | Mar. | 0.0024 | 0.0881 | 0.1488 |
| | Apr. | 0.4735 | 0.8535 | 0.9471 |
| | May | 0.0294 | 0.2742 | 0.4290 |
| | Jun. | 0.8535 | 0.8535 | 0.9471 |
| | Jul. | 0.3999 | 0.8535 | 0.9471 |
| | Aug. | 0.6550 | 0.8535 | 0.9471 |
| | Sep. | 0.5917 | 0.8535 | 0.9471 |
| | Oct. | 0.8466 | 0.8535 | 0.9471 |
| | Nov. | 0.0217 | 0.2692 | 0.4290 |
| | Dec. | 0.0050 | 0.0922 | 0.1558 |
| Day of Week | Monday | 0.3075 | 0.9471 | 0.9471 |
| | Tuesday | 0.0341 | 0.3893 | 0.4290 |
| | Wednesday | 0.2415 | 0.9471 | 0.9471 |
| | Thursday | 0.9471 | 0.9471 | 0.9471 |
| | Friday | 0.5054 | 0.9471 | 0.9471 |


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- Research conclusions depend on what kind of hypotheses are proposed, how explanatory variables are chosen, how data is processed, how data is analyzed, and how results are interpreted.
- However, all of these can vary depending on empirical settings:
  - Should data from the past 20 years or the past 10 years be used?
  - Should factors be extracted from the entire market, index constituents, or industries?
  - Should factors be constructed using portfolio sort or cross-sectional regression (i.e., pure factors)?
  - How should outliers be handled? What about normalization? What about winsorization?
  - ...


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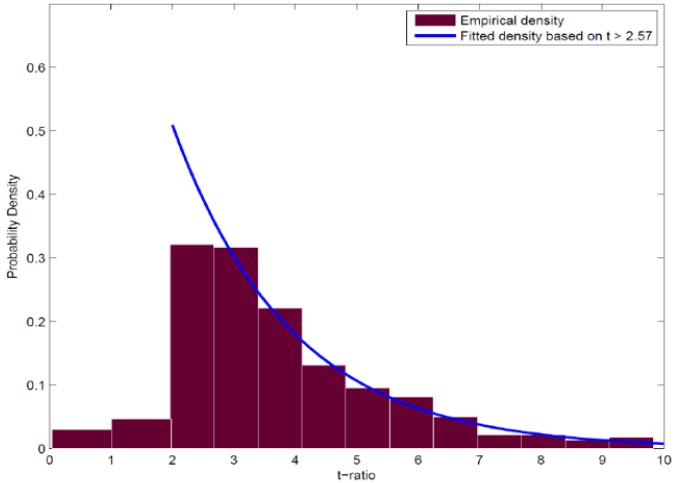
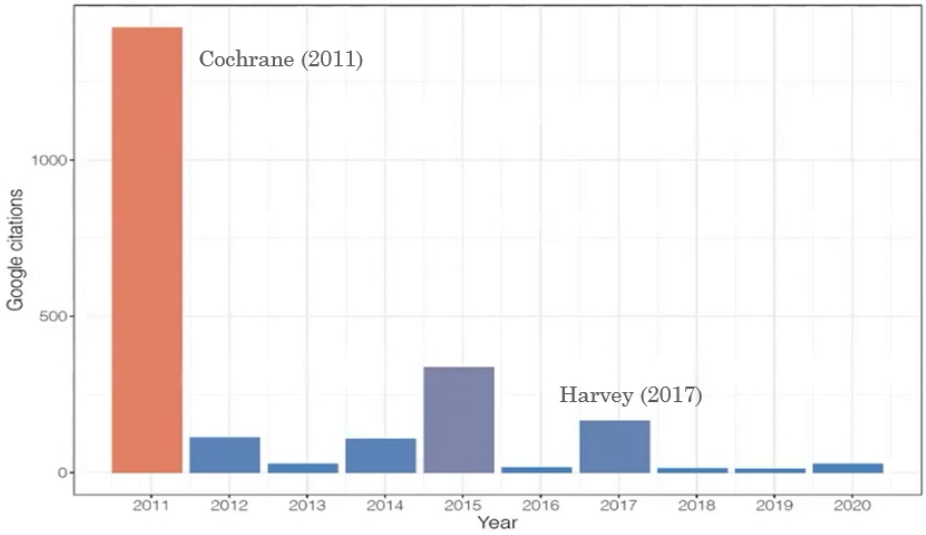



**Figure 1. Distribution of reported $t$-statistics from factor studies, 1963 to 2012.** Evidence from

Source: Harvey et al. (2016)


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- AFA presidential address citations:





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# Mismatch Between Environment and Genes



Type I error



Type II error

- For animals to survive, they must minimize Type II error, even if it means allowing a higher rate of Type I error (false discovery).
- This tendency has been passed down through generations during evolution. Humans tend to allow a higher rate of Type I error when analyzing problems, making overfitting more likely to occur.


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- **Frequentist methods:** This approach aims to **control Type I errors by increasing the standard errors**, thereby lowering the significance level of individual factors.
  - Increasing the standard errors means expanding the confidence intervals, thus making the significance threshold for individual tests more stringent.
  - Only factors with original $t$-statistics far exceeding the traditional level of 2.0 are likely to remain significant after the adjustment.

- **Bayesian methods:** This approach aims to calculate the **posterior** probability of a factor being true. Its advantage lies in the ability to incorporate appropriate financial priors within the Bayesian framework, but its downside is the complexity of computation, often requiring compromises and simplifications.


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# Two Research/Solution Streams

## Frequentist

Family-wise error rate

- Bonferroni (1936)
- Holm (1979)
- White (2000)
- Romano and Wolf (2005, 2007)

False discovery rate

- Benjamini and Hochberg (1995)
- Benjamini and Yekutieli (2001)
- Harvey, Liu and Zhu (2016)

False discovery proportion

- Romano and Wolf (2007)
- Romano, Shaikh and Wolf (2008)
- Chordia, Goyal and Saretto (2020)

## Bayesian

Full Bayesian

- Scott and Berger (2006)
- Harvey, Liu and Zhu (2016)

Minimum Bayesian factor

- Harvey (2017)

Bayesian extension

- Harvey and Liu (2020, 2021)


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# Error Table

- The goal of frequentist methods is to control Type I errors: family-wise error rate (FWER), the false discovery rate (FDR), and the false discovery proportion (FDP) are all Type I error type metrics.

Table: Error Table

| | $\mathcal{H}_0$ Accepted | $\mathcal{H}_0$ Rejected | Total |
|---|---|---|---|
| $\mathcal{H}_0$ is True | $T_0$ | $F_1$ | $K_0$ |
| $\mathcal{H}_0$ is False | $F_0$ | $T_1$ | $K_1$ |
| Total | $K - L$ | $L = F_1 + T_1$ | $K$ |

- Suppose there are $K$ factors being tested, with $K_0$ true null hypotheses and $K_1$ false null hypotheses. At a given significance level (e.g., 0.05), $L$ null hypotheses are rejected, including $T_1$ true discoveries and $F_1$ false discoveries. We can then define the FWER, FDR, and FDP using $F_1$ and $L$.


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# Family-Wise Error Rate (FWER)



- FWER is defined as the probability of **at least one false discovery**, i.e., prob$(F_1 \geq 1)$. At a given significance level $\alpha_s$, the mathematical expression for controlling the FWER is:

$$\text{prob}(F_1 \geq 1) \leq \alpha_s.$$

- Controlling the FWER is quite **stringent** for individual hypotheses, greatly increasing Type II errors. (Type II error refers to the scenario in hypothesis testing where the null hypothesis $\mathcal{H}_0$ is false but is incorrectly accepted.)

- The earliest algorithms for controlling the FWER include the Bonferroni correction and the Holm correction. Both methods **lift the $p$-values** of individual tests to control the FWER.


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- Bonferroni correction for the original $p$-values goes as follows:

$$p_i^{\text{Bonferroni}} = \min\{Kp_i, 1\}, \quad i = 1, \cdots, K,$$

  where $p_i$ and $p_i^{\text{Bonferroni}}$ represent the original and adjusted $p$-values of the $i$-th factor, respectively. $K$ is the number of factors (i.e., hypotheses being tested.)

- Suppose there are six factors with original $p$-values: 0.005, 0.009, 0.0128, 0.0135, 0.045, and 0.06. Without considering the multiple hypothesis testing issue, at a significance level of 0.05, the first five factors would be deemed significant.

- Applying Bonferroni correction, new $p$-values are: 0.03, 0.054, 0.0768, 0.081, 0.27, and 0.36; only the first factor remains significant at the 0.05 significance level.

- Holm correction adjusts the original $p$-values **sequentially** from smallest to largest:

$$p_i^{\text{Holm}} = \min\left\{\max_{j\le i}\{(K-j+1)p_j\}, 1\right\}, \quad i = 1, \cdots, K.$$

- For example:

$$\begin{aligned}
p_1^{\text{Holm}} &= \max\{6p_1\} = 0.03,\\
p_2^{\text{Holm}} &= \max\{6p_1, 5p_2\} = 5p_2 = 0.045,\\
p_3^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3\} = 4p_3 = 0.0512,\\
p_4^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3, 3p_4\} = 4p_3 = 0.0512,\\
p_5^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3, 3p_4, 2p_5\} = 2p_5 = 0.09,\\
p_6^{\text{Holm}} &= \max\{6p_1, 5p_2, 4p_3, 3p_4, 2p_5, p_6\} = 2p_5 = 0.09.
\end{aligned}$$

Only the first two factors remain significant at the 0.05 significance level.


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- New algorithms: bootstrap reality check method of White (2000), StepM and $k$-StepM methods of Romano and Wolf (2005, 2007). These three algorithms use **bootstrap** to resample the factor return data and combine it with **orthogonalization** to determine the cutoff for the $t$-statistics, thus avoiding any assumptions about the data distribution.

- **Orthogonalization** eliminates the sample mean of the factor returns, making the factor returns zero-mean random disturbances over time.

- **Bootstrap** samples the orthogonalized returns to derive the distribution of test statistics caused purely by chance (noise), thereby determining whether the significance of the original factors is genuine or merely noise.

- Due to their stringent nature, methods targeting the control of the family-wise error rate are not well-suited for the financial field.


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- Setup:
  - $K$ factors being tested, and a total period of $T$,
  - $T \times K$ matrix $\mathbf{D}$: time series of factor returns.
- Bootstrap (let me explain this on whiteboard):
  - For each bootstrap sample $\mathbf{B}^b$, calculate the bootstrap $t$-statistic for each factor:

$$t_k^b = \frac{\bar{\mu}_k^b - \bar{\mu}_k}{\sigma_k^b},$$

    where $\bar{\mu}_k$ is the average return of factor $k$ in the **original sample**, $\bar{\mu}_k^b$ is the average return of factor $k$ in the **bootstrap sample** $b$, and $\sigma_k^b$ is the standard error of $\bar{\mu}_k^b$.
- Input of Algorithm:
  - For each bootstrap sample, we derive a vector of bootstrap $t$-statistics.
  - For a total of $B$ bootstraps, we derive a $K \times B$ matrix $\mathbf{Z}$ of bootstrap $t$-statistics.


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# Bootstrap Reality Check: The Algorithm



The goal of bootstrap reality check is to test whether the top-ranked factor (i.e., the 1st) in terms original $t$-statistic remains significant after accounting for multiple hypothesis testing.

1. For each column of $\mathbf{Z}$ (i.e., the bootstrap $t$-statistics of $K$ factors from a given bootstrap sample), take the absolute values and find the maximum.

2. From the total of $B$ maximum values obtained from step 1, find the $1 - \alpha_s$ quantile, which services as the cutoff for the $t$-statistic obtained purely by chance given the significance level $\alpha_s$.

3. Compare the original highest $|t$-statistic$|$ against the cutoff. If it exceeds the cutoff, then the corresponding factor is considered significant at significance level $\alpha_s$.


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# False Discovery Rate (FDR)



- FDR is defined as the expected value of $F_1/L$, i.e., FDR $\equiv \mathbb{E}[F_1/L]$.
- Controlling it at a given level $\delta$ can be expressed as:

$$\mathbb{E}[F_1/L] \leq \delta.$$

- **Controlling the FDR allows $F_1$ to increase with $L$.** When the FDR is used instead of FWER as the control target, more null hypotheses are rejected.
- Representative method is the Benjamini-Hochberg-Yekutieli (BHY) correction.


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- BHY correction starts with the **largest** original $p$-value and adjusts in reverse order, as follows:

$$p_i^{\text{BHY}} = \begin{cases} p_K & \text{if} \quad i = K, \\ \min\left\{ p_{i+1}^{\text{BHY}}, \dfrac{K \times c(K)}{i} p_i \right\} & \text{if} \quad i \leq K-1, \end{cases}$$

  where $c(K) = \sum_{j=1}^{K} \frac{1}{j}$, and $p_{i+1}^{\text{BHY}}$ is the adjusted $p$-value.

- Recall the previous example: 0.005, 0.009, 0.0128, 0.0135, 0.045, and 0.06. Since $K = 6$, $c(K) = 2.45$.

- BHY-adjusted $p$-values: 0.0496, 0.0496, 0.0496, 0.0496, 0.06, and 0.06. At the 0.05 significance level, the first four factors remain significant.

- **Compared to the FWER adjustments, more factors remain significant when the BHY adjustment is applied.**


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# False Discovery Proportion (FDP)



- FDP is defined as the probability that $F_1/L$ exceeds a given threshold $\gamma$, i.e., $\text{prob}(F_1/L \geq \gamma)$.
- Controlling it under a given significance level $\alpha_s$ means:

$$\text{prob}(F_1/L \geq \gamma) \leq \alpha_s.$$

- It is similar to the false discovery rate. In finance, it is generally considered better to target the control of the false discovery rate and the false discovery proportion, over family-wise error rate.


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- Harvey et al. (2016) raised the $t$-statistic cutoff to 3.0.
- However, when attempting to eliminate the impact of multiple hypothesis testing, the **the base number** of null hypotheses being tested simultaneously (i.e., number of factors being evaluated) is an important contributor.
- The base number determines the amount of luck. For example, testing 100 factors versus 10,000 factors makes a significant difference; the best one from the latter is undoubtedly more significant than the best one from the former.
- Question: Do those published factors represent the total number of factors studied by academia? **NO!**
- We have no idea about how many additional factors were tried behind the scenes. Since the published factors are a subset of all researched factors, we can only infer that **the 3.0 cutoff is conservative.**


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# Chordia, Goyal and Saretto (2020)

## 3 The "Universe" Issue

- Let $\mathcal{P}$ represent the set of published factors and $\mathcal{R}$ represent the set of all factors studied. Assume $\mathcal{P} = \{s \in \mathcal{R}, t_s \geq 2.0\}$. Since $\mathcal{P}$ is only a subset (lower bound) of $\mathcal{R}$, using $\mathcal{P}$ to control Type I errors inevitably underestimates the effect of luck.

- The authors constructed 2,393,641 factors through simple arithmetic operations using accounting data. Let the set $\mathcal{E}$ represent these factors, forming the upper bound of $\mathcal{R}$. (However, the number of factors researched by **finance** scholars should be (far) fewer than 2,393,641.)

- The authors used simulation to uncover factor publication process and calibrated the model parameters using real U.S. stock market data.

- They raised the $t$-statistic cutoff to over 3.4.


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# Comments



Two common issues with frequentist approach:

- Ignoring priors, assuming that the risk premium of **all** factors being tested are zero (also known as ensemble null prior). This assumption is very problematic from an industry perspective. (Why? Let me explain.)

- $t$-cutoff is determined by pre-specified Type I error rate, without considering the trade-off between Type I and Type II errors.

Type II error means that a factor is capable of generating excess returns, but the null is accepted. Frequentist approach often has a high Type II error rate, resulting a lower **test power** (test power = 1− Type II error rate).


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- The first hierarchy of this model is the **prior distribution** of sample average return $X_i$:

$$(X_i|\mu_i, \sigma^2, \gamma_i) \sim \mathcal{N}(\gamma_i\mu_i, \sigma^2),$$

  where:
  - $\mu_i$ represents the mean,
  - $\sigma^2$ represents the variance (the variance is not subscripted by $i$, implying the assumption that the variances of average returns for all factors are the same),
  - $\gamma_i$ is a binary variable, with 1 indicating the factor is genuine and 0 indicating the factor is false.

- The critical and stringent assumption of this prior is that the factor returns are **conditionally independent**. This is important to derive the likelihood function.


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- Let $\mathbf{X}$, $\boldsymbol{\mu}$, and $\boldsymbol{\gamma}$ represent the vectors corresponding to $X_i$, $\mu_i$, and $\gamma_i$, respectively.
- Under the independence assumption, the likelihood function is:

$$f(\mathbf{X}|\sigma^2, \boldsymbol{\mu}, \boldsymbol{\gamma}) = \prod_i \left[ \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left( \frac{-(X_i - \gamma_i \mu_i)^2}{2\sigma^2} \right) \right].$$

- The first hierarchy of the model describes the distribution of the average returns of factors given $\mu_i$ and $\gamma_i$, but it does not specify how $\mu_i$ and $\gamma_i$ are derived.


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- The second hierarchy specifies the priors for $\mu_i$ and $\gamma_i$:

$$\begin{aligned} \mu_i|\tau^2 &\overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \\ \gamma_i|p_0 &\overset{\text{iid}}{\sim} \text{Ber}(1 - p_0). \end{aligned}$$

- $\mu_i$ follows an iid normal distribution (with a mean of zero and a variance of $\tau^2$).
- $\gamma_i$ follows a Bernoulli distribution with parameter $p_0$, meaning that the probability of $\gamma_i = 0$ is $p_0$ and the probability of $\gamma_i = 1$ is $1 - p_0$.


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- The third hierarchy specifies the prior distributions for $\tau$, $\sigma$, and $p_0$:

$$\begin{aligned} (\tau^2, \sigma^2) &\sim \pi_1(\tau^2, \sigma^2), \\ p_0 &\sim \pi_2(p_0). \end{aligned}$$

- In the absence of sufficient prior knowledge, Scott and Berger (2006) recommend using the following prior distributions:

$$\begin{aligned} \pi_1(\tau^2, \sigma^2) &\propto (\tau^2 + \sigma^2)^{-2}, \\ \pi_2(p_0) &= (1+\alpha) p_0^{\alpha}. \end{aligned}$$

- In $\pi_2$, $\alpha$ controls the center of its distribution (for example, when $\alpha = 0$, $\pi_2$ becomes a uniform distribution). $p_0$ controls the probability that each factor is false.

- Common sense suggests that most factors should be false, so the value of $p_0$ should be close to 1. Additionally, as the number of factors considered simultaneously increases, $\alpha$ can be adjusted to further control the distribution of $p_0$ (making it closer to 1).


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# Posterior Probability



- With the full Bayesian framework, one can calculate the **posterior probability** of each factor being true.
- The formula for calculating the posterior probability $p_i \equiv \text{prob}(\gamma_i = 1|\mathbf{X})$ is very complex. Check Scott and Berger (2006) if you are interested.
- The posterior probability decreases with the number of hypotheses tested simultaneously. **As the number of false factors increases, the evidence conveyed by true factors will diminish, reflecting a penalty for multiple hypothesis testing corresponding to frequentist methods.**
- Although theoretically sound, the approach has many practical issues:
  - Conditional independence are too stringent.
  - (Computational) When considering a large number of factors, calculating the posterior probability of each factor being true is extremely challenging.
  - Lack of decision criterion, i.e., what's the cutoff value for posterior probability?


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# Presidential Address: The Scientific Outlook in Financial Economics

CAMPBELL R. HARVEY*

**ABSTRACT**

Given the competition for top journal space, there is an incentive to produce "significant" results. With the combination of unreported tests, lack of adjustment for multiple tests, and direct and indirect $p$-hacking, many of the results being published will fail to hold up in the future. In addition, there are basic issues with the interpretation of statistical significance. Increasing thresholds may be necessary, but still may not be sufficient: if the effect being studied is rare, even $t > 3$ will produce a large number of false positives. Here I explore the meaning and limitations of a $p$-value. I offer a simple alternative (the minimum Bayes factor). I present guidelines for a robust, transparent research culture in financial economics. Finally, I offer some thoughts on the importance of risk-taking (from the perspective of authors and editors) to advance our field.


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- By definition, a $p$-value represents the conditional probability of observing a certain (extreme) event under the null hypothesis. Therefore, a lower $p$-value suggests that the average return observed in the sample data is less likely under the null hypothesis (i.e., the expected return of the factor is zero).

- If we denote an extreme event by $D$ (e.g., the $t$-statistic exceeding a certain value), the $p$-value represents $\text{prob}(D|\mathcal{H}_0)$.

**However, is "a lower $p$-value" equivalent to "the factor is more likely to generate excess returns"?**


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- For the latter, we care about **the conditional probability that the null hypothesis is true given the occurrence of the extreme event** $D$, i.e., prob$(\mathcal{H}_0|D)$, but this is not the $p$-value.

**Confusing $p$-value $\equiv$ prob$(D|\mathcal{H}_0)$ with prob$(\mathcal{H}_0|D)$ is simply a mistake!**

- As a side note, the American Statistical Association has long warned against the misuse of $p$-values, and many top economics journals (such as *Econometrica* and *American Economic Review*) have included policies discouraging the use of stars, i.e., $\star$, $\star\star$, $\star\star\star$, to indicate statistical significance.

- In 2019, the official journal of the American Statistical Association, *The American Statistician*, published a special issue titled "Statistical Inference in the 21st Century: A World Beyond p $< 0.05$", with over 40 articles criticizing the misuse of $p$-values and providing feasible alternatives. In this special issue, the American Statistical Association even recommended banning the term *statistically significant*.

- **Prior Odds Ratio:** the ratio of the prior probabilities of two competing hypotheses $H_0$ (null) and $\mathcal{H}_1$ (alterantive):

$$\text{Prior Odds Ratio} = \frac{\pi_0}{\pi_1}.$$

- **Posterior Odds Ratio:** the ratio of the posterior probabilities of the same hypotheses, given the observed data $D$:

$$\text{Posterior Odds Ratio} = \frac{a_0}{a_1}.$$

- **Bayes Factor:** the ratio between the two:

$$\text{Bayes Factor} = \frac{\text{Posterior Odds Ratio}}{\text{Prior Odds Ratio}} = \frac{a_0/a_1}{\pi_0/\pi_1}.$$


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- Consider the simplest case where both $\mathcal{H}_0$ and $\mathcal{H}_1$ are simple hypotheses:

$$\mathcal{H}_0 : \Theta_0 = \theta_0, \quad \mathcal{H}_1 : \Theta_1 = \theta_1.$$

- In this case:

$$a_0 = \text{prob}(\Theta_0|\text{data}) = \frac{f(\boldsymbol{x}|\theta_0)\pi_0}{f(\text{data}|\theta_0)\pi_0 + f(\text{data}|\theta_1)\pi_1}, \quad a_1 = \text{prob}(\Theta_1|\boldsymbol{x}) = \frac{f(\text{data}|\theta_1)\pi_1}{f(\text{data}|\theta_0)\pi_0 + f(\text{data}|\theta_1)\pi_1}.$$

- The posterior odds ratio becomes:

$$\frac{a_0}{a_1} = \frac{f(\text{data}|\theta_0)\pi_0}{f(\text{data}|\theta_1)\pi_1}.$$

- The Bayes factor is:

$$\text{Bayes Factor} = \frac{a_0/a_1}{\pi_0/\pi_1} = \frac{f(\text{data}|\theta_0)}{f(\text{data}|\theta_1)}.$$

It is the **likelihood ratio** of the observed data under the two hypotheses.


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- Back to our context, when testing a factor's expected return, the null hypothesis is usually that it is zero, so $\mathcal{H}_0$ can be written as $\Theta_0 = 0$.
- For the alternative hypothesis, to make the analysis more general, it is often assumed that under $\mathcal{H}_1$, the corresponding parameter $\theta_1$ follows some prior distribution $\pi_A(\theta_1)$.
- The likelihood function is $\int f(\text{data}|\theta_1)\pi_A(\theta_1)d\theta_1$, and the Bayes factor becomes:

$$\text{Bayes Factor} = \frac{f(\text{data}|\theta_0)}{\int f(\text{data}|\theta_1)\pi_A(\theta_1)d\theta_1}.$$


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- For testing factors, the **posterior odds ratio** is what we are truly concerned with.

**A low posterior odds ratio indicates a low posterior probability of the null hypothesis, allowing us to safely reject it, meaning that the factor is likely genuine.**

- Calculating the bayes factor requires specifying the prior distribution under the alternative hypothesis, which is often very difficult.

- However, among the various possible values of the Bayes factor, there is a special value, known as the **minimum Bayes factor** (MBF).


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- Recall that the three concepts are related by:

$$\text{Posterior Odds Ratio} = \text{Prior Odds Ratio} \times \text{Bayes Factor}.$$

- This equation can be understood as follows: starting with the prior odds ratio for $\mathcal{H}_0$ and $\mathcal{H}_1$, and then adjusting it by multiplying the Bayes factor to obtain the posterior odds ratio.

**It reflects how the initial beliefs (the prior odds) are updated by the evidence provided by the data (through the Bayes factor) to form the final beliefs (the posterior odds).**

$$\text{Posterior Odds Ratio} = \text{Prior Odds Ratio} \times \text{Bayes Factor}.$$

- The **smaller** the Bayes factor (and thus the lower the posterior odds ratio), the **weaker** our belief in the null hypothesis becomes in the posterior compared to the prior;
- The **larger** the Bayes factor (and thus the higher the posterior odds ratio), the **stronger** our belief in the null hypothesis becomes in the posterior compared to the prior.

**Bayes factor measures the extent to which we deviate from the prior odds ratio after observing sample data, and the minimum Bayes factor provides the strongest deviation against the null hypothesis.**


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- MBF corresponds to a special prior distribution under the alternative hypothesis that provides the strongest evidence **in favor of the alternative hypothesis** (or the strongest evidence **against the null hypothesis**).
- Illustration: Suppose there are 1,000 observations of factor returns, with a sample mean of 4%. Under what circumstances would we obtain the MBF?
- Answer: When the prior distribution under the alternative hypothesis is concentrated at the sample mean of 4%.

**When the density of the prior distribution under the alternative hypothesis is concentrated at the maximum likelihood estimate of the data, the Bayes factor is minimized.**


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- Harvey (2017) explained that using original $p$-value or $t$-statistic, MBF can be calculated as follows:

$$\begin{aligned} \text{MBF} &= -\exp(1) \times p\text{-value} \times \ln(p\text{-value}), \\ \text{MBF} &= \exp(-t\text{-statistic}^2/2). \end{aligned}$$

- We can derive the posterior probability of the null hypothesis being true:

$$\begin{aligned} \text{Posterior Prob. of Null} &= \frac{\text{Posterior Odds Ratio}}{1 + \text{Posterior Odds Ratio}} \\ &= \frac{\text{MBF} \times \text{Prior Odds Ratio}}{1 + \text{MBF} \times \text{Prior Odds Ratio}}. \end{aligned}$$

- Harvey termed it **Bayesian posterior $p$-value**. It addresses the question that people truly care about, namely, the conditional probability that the null hypothesis is true given the data.


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- Little economic rationale: 49:1
- Plausible factors: 4:1
- Strong economic theory support: 1:1

| Factor | Prior Odds Ratio | Posterior $p$-value | Interpretation |
|---|---|---|---|
| Clever Tickers | 49:1 | 0.588 | Unlikely to generate excess returns |
| Profitability | 4:1 | 0.053 | Quite likely to be a genuine factor |
| Market | 1:1 | 0.035 | Undoubtedly a genuine factor |

Table: Comparison of Factors and Bayesian Posterior $p$-Values


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6 Extension of Bayesian Approach


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- Some latest research findings from Harvey and Liu control the proportion of true factors through a prior parameter $p_0$. They can be seen as extensions of the Bayesian approach.

- By introducing $p_0$, these methods not only make the handling of multiple hypothesis testing more realistic but also allow for a trade-off between Type I and Type II errors.

**In the context of factor discovering, a Type II error occurs when a factor is truly valid but is incorrectly deemed to have no excess returns.**


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# A Double-Bootstrap Method



- Harvey and Liu (2020) propose a double-bootstrap method.
- Instead of ensemble null assumption, they adopt the bi-modal mean distribution:
  - Most factors being false, with their expected returns coming from a distribution with a mean of zero,
  - A small portion of factors are genuine, with their expected returns coming from a distribution with a non-zero mean.
- One can choose the value of $p_0$ based on their own experience (i.e., prior knowledge), which reflects the Bayesian idea.
- Once $p_0$ is selected, a natural thought is that the top $p_0K$ factors with the highest $t$-statistics among all $K$ factors are genuine. However, due to sampling uncertainty, it may not be the case.


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- Let $T \times K$ matrix $\mathbf{X}_0$ represent the time series of factor returns. The first hierarchy of bootstrap sampling is applied to $\mathbf{X}_0$. (Do you still remember how?)

- Each time a bootstrap sample is drawn, a $T \times K$ matrix $\mathbf{X}_i$ is obtained (where the subscript $i$ denotes the $i$-th bootstrap sample). For $\mathbf{X}_i$, the $t$-statistic of each factor is calculated, and the top $p_0 K$ factors with the highest $t$-statistics are considered to be genuine for that bootstrap sample.

- In $\mathbf{X}_0$, the average returns of these factors are replaced with their means in $\mathbf{X}_i$, and the returns of the remaining factors in $\mathbf{X}_0$ are demeaned (i.e., treated as random noise), thereby constructing a $T \times K$ matrix $\mathbf{Y}_i$. (Demeaning the returns of the remaining $(1 - p_0)K$ factors over time embodies the concept of *orthogonalization*.)


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- The $\mathbf{Y}_i$ matrices obtained from the first-stage bootstrap sampling serve as the *original* data for the second-stage bootstrap.
- That is, $\mathbf{Y}_i$ is considered the *population* for the second-stage bootstrap. (For each $\mathbf{Y}_i$, as we know which factors are genuine and which are false, the population is known to us.)
- By performing bootstrap sampling on $\mathbf{Y}_i$, we can construct the error table, making it easy to calculate both Type I and Type II error rates.
- For the $j$-th bootstrap sample result of $\mathbf{Y}_i$, define the following four variables using the error table:
  - $\mathrm{TN}^{i,j}$: The number of null hypotheses correctly accepted (true negative).
  - $\mathrm{TP}^{i,j}$: The number of null hypotheses correctly rejected (true positive).
  - $\mathrm{FP}^{i,j}$: The number of null hypotheses incorrectly rejected (false positive).
  - $\mathrm{FN}^{i,j}$: The number of null hypotheses incorrectly accepted (false negative).


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- To categorize the hypothesis test results of all factors in each bootstrap sample of $\mathbf{Y}_i$ into the four categories above, we need to specify a $t$-statistic cutoff.

> **Example:** Suppose that factor A is genunie in $\mathbf{Y}_i$. Further, assume that in the $j$-th bootstrap sample of $\mathbf{Y}_i$, the $t$-statistic of factor A is 2.0. Now, suppose the pre-selected $t$-statistic cutoff is 2.3. Since $2.0 < 2.3$, the null hypothesis for factor A would be accepted. However, we know that factor A is genuine in $\mathbf{Y}_i$, meaning the null hypothesis is false. Therefore, it is incorrectly accepted, so it is categorized as $\text{FN}^{i,j}$.

- $\text{TN}^{i,j}$, $\text{TP}^{i,j}$, $\text{FP}^{i,j}$, and $\text{FN}^{i,j}$ are functions of the $t$-statistic cutoff.
- Using these four variables, we can define Realized False Discovery Rate (RFDR), the Realized Rate of Miss (RMISS), and the Realized Ratio of False Discoveries to Misses (RRATIO). They are also all functions of the cutoff!


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- Realized False Discovery Rate (RFDR):

$$\text{RFDR}^{i,j} = \begin{cases} \dfrac{\text{FP}^{i,j}}{\text{FP}^{i,j} + \text{TP}^{i,j}}, & \text{if} \quad \text{FP}^{i,j} + \text{TP}^{i,j} > 0, \\ 0, & \text{if} \quad \text{FP}^{i,j} + \text{TP}^{i,j} = 0. \end{cases}$$

- Realized Rate of Miss (RMISS):

$$\text{RMISS}^{i,j} = \begin{cases} \dfrac{\text{FN}^{i,j}}{\text{FN}^{i,j} + \text{TN}^{i,j}}, & \text{if} \quad \text{FN}^{i,j} + \text{TN}^{i,j} > 0, \\ 0, & \text{if} \quad \text{FN}^{i,j} + \text{TN}^{i,j} = 0. \end{cases}$$

- Realized Ratio of False Discoveries to Misses (RRATIO):

$$\text{RRATIO}^{i,j} = \begin{cases} \dfrac{\text{FP}^{i,j}}{\text{FN}^{i,j}}, & \text{if} \quad \text{FN}^{i,j} > 0, \\ 0, & \text{if} \quad \text{FN}^{i,j} = 0. \end{cases}$$


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- In practice, a total of $J$ bootstrap samples are drawn for each $\mathbf{Y}_i$. Additionally, suppose that $I$ bootstrap samples $\mathbf{Y}_i$ were generated from the original data $\mathbf{X}_0$ in the first-stage bootstrap.

- For each of $\text{TN}^{i,j}$, $\text{TP}^{i,j}$, $\text{FP}^{i,j}$, and $\text{FN}^{i,j}$, the double-bootstrap method produces $I \times J$ values. Taking the average gives the Type I error rate (TYPE1), Type II error rate (TYPE2), and the ratio of the two errors (ORATIO):

$$\begin{aligned}
\text{TYPE1} &= \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J}\text{RFDR}^{i,j}, \\
\text{TYPE2} &= \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J}\text{RMISS}^{i,j}, \\
\text{ORATIO} &= \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J}\text{RRATIO}^{i,j}.
\end{aligned}$$


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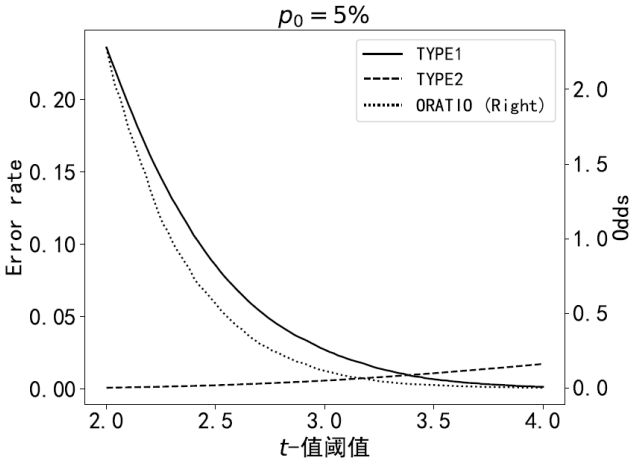


# Comments

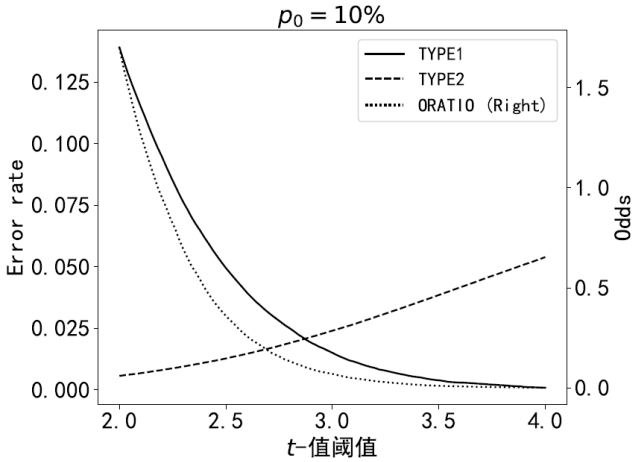


- These three final indicators are also functions of the $t$-statistic cutoff.
- TYPE1 decreases as the $t$-statistic threshold increases, while TYPE2 increases with the the threshold, highlighting the trade-off between the two. This allows users to select the most appropriate $t$-statistic cutoff based on the issue they are most concerned with.
- The double-bootstrap method ensures that the $t$-statistic cutoff determined also corresponds to the optimal Type II error rate. (i.e., it has higher power.)


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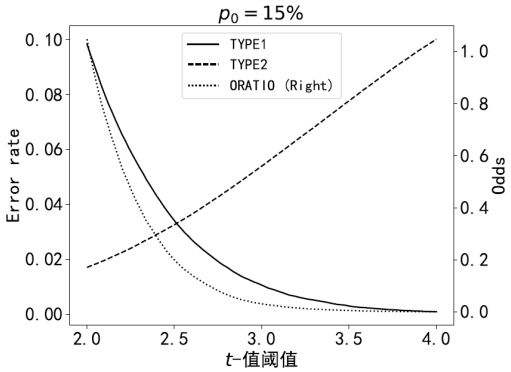



(a) $p_0 = 5\%$

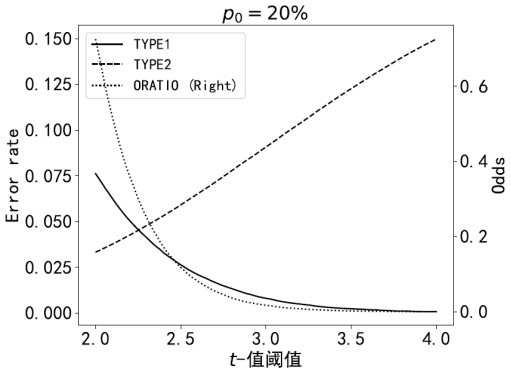



(b) $p_0 = 10\%$


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(c) $p_0 = 15\%$

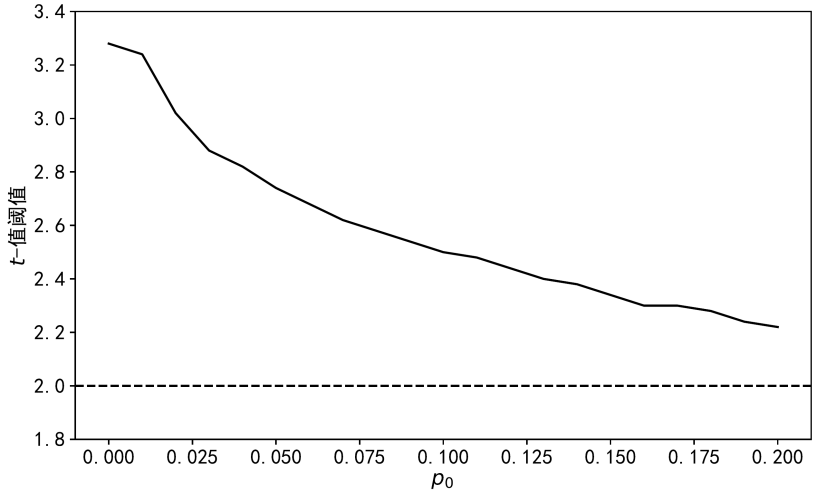



(d) $p_0 = 20\%$

- When $p_0$ is high, the Type I error rate decreases rapidly as the $t$-statistic cut increases, while the Type II error rate climbs quickly. If the chosen $t$-statistic cutoff is too high (i.e., too stringent), it is likely that genuine factors will be missed, resulting in a Type II error.


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- $t$-statistic cutoffs required to control the Type I error rate at 5% for different values of $p_0$.
- An appropriate $p_0$ can yield a more realistic cutoff, minimizing the Type II error rate at a given Type I error rate level.





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# Debate

## 7 Criticism and Response

- Numerous empirical results from Harvey and his co-authors not only challenge a significant portion of the factors unearthed in academic research over the past few decades but also, to some extent, question the authority of academic research itself.
- To protect the integrity of academic work and research outcomes, voices of dissent have emerged from the other side of the debate as well.

> The academic community (especially top-tier journals) is not inclined to admit that many empirical asset pricing articles published in prestigious journals may contain false discoveries.

- The critique by Chen (2021) is particularly notable. It is published in *Journal of Finance*, the official publication of the American Finance Association. This reflects the mainstream academic community's skepticism regarding the existence of (a large number of) false discoveries.


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# A Thought Experiment



- Chen (2021) uses the ensemble null prior.
- Each factor has a certain probability of being observed (i.e., published) depending on its $t$-statistic.
- With this setup, the paper's most important formula is the expected number of attempted factors:

$$\mathbb{E}[K_0] \geq \frac{\mathbb{E}[\text{num observed } (|t_i| > \bar{t})]}{\text{prob}(|Z| > \bar{t})},$$

  where $\bar{t}$ is the selected $t$-statistic cutoff.
- This formula indicates the lower bound on the number of factors that need to be explored to observe $\mathbb{E}[\text{num observed } (|t_i| > \bar{t})]$ factors with $t$-statistics exceeding $\bar{t}$.


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# Is It Even Valid?

7 Criticism and Response

- Among the 210 factors considered in the paper, 26 have $t$-statistics over 6.0, and the calculations suggest that the academic community would need to investigate at least **13 billion factors** to identify 26 factors with such high levels of significance.
- Even if all the world's finance professors worked nonstop to mine factors, it would take **451 years.**
- The ratio of unpublished to published factors is as high as $10^{14} : 1$.
- Conclusion: $p$-hacking alone cannot explain the many highly significant factors discovered in academia. **Of course not! But we knew it! No one denies it!**

**Don't these shocking conclusions precisely indicate that the ensemble null prior itself is questionable?**


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- Harvey and Liu simulate the factor publication process:

$$\text{prob}(t\text{-statistic}) = \frac{\gamma_1}{2\gamma_1 + \exp(-\gamma_2(t\text{-statistic} - 2.0))},$$

  where $\gamma_1$ and $\gamma_2$ are positive parameters.

- The probability increases monotonically with a factor's $t$-statistic, and approaches 0.5 as $t$-statistic goes to infinity. (Reasonable. Why?)

- This model differs in the following important aspects:
  1. It considers all factors that have been attempted through $K_0$.
  2. It replaces the ensemble null prior with a bi-modal mean prior through $p_0$.
  3. It describes the factor publication process using parameters $\gamma_1$ and $\gamma_2$.


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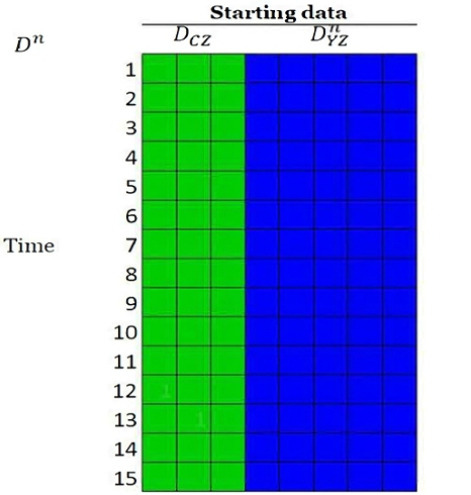
- This model has the following set of parameters $[K_0, p_0, \gamma_1, \gamma_2]$.
- It simulates the factor publication process and its parameters will be calibrated based on real data from published factors.
- Pool: Since $K_0$ represents all attempted anomalies (including both published and unpublished ones), a large pool of factors is required, from which $K_0$ factors will be drawn.

> Harvey and Liu create a combined pool by utilizing both the 156 published factors mentioned in Chen and Zimmermann (2020) and the 18,000+ factors generated purely through data-mining by Yan and Zheng (2017).


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- Draw $K_0$ factors from this combined pool. (Each simulation needs to contain those 156 published factors; the remaining $K_0 - 156$ factors are randomly selected from the pool of purely data-mined factors.)

- $T \times 156$ matrix $\mathbf{D}_{CZ}$ represent the return series of the published factors, and the $T \times (K_0 - 156)$ matrix $\mathbf{D}^n_{YZ}$ represent the return series of the remaining factors. $T \times K_0$ matrix $\mathbf{D}^i = [\mathbf{D}_{CZ}, \mathbf{D}^i_{YZ}]$ represents the return series of the $i$-th simulation.

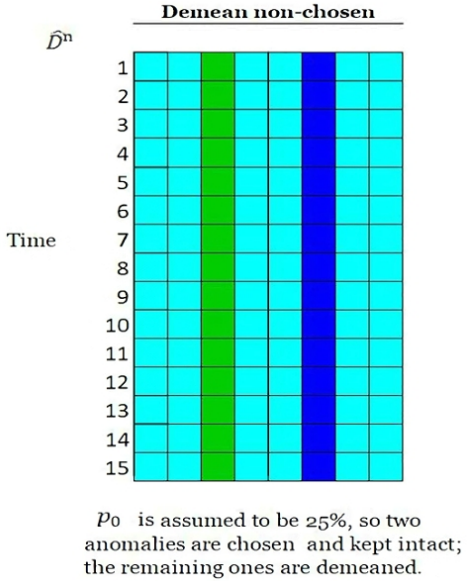



For simplicity, we assume CZ include 3 anomalies and 5 anomalies are randomly chosen from YZ.


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# Step 2: Demean

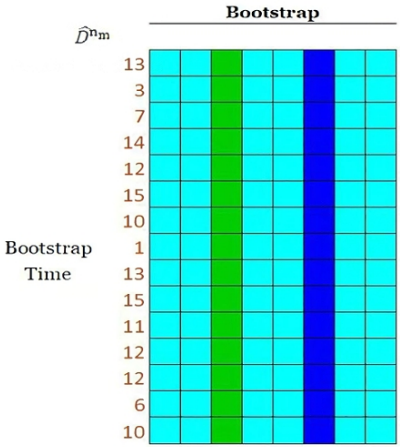
- Calculate the absolute values of the $t$-statistics of all factors using $\mathbf{D}^i$
- Retain the $p_0 K_0$ factors with the highest absolute values, and mean-adjust the returns of the remaining $(1 - p_0)K_0$ factors over time.
- The resulting $T \times K_0$ return series matrix is denoted as $\hat{\mathbf{D}}^i$.





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- Treat $\hat{\mathbf{D}}^i$ as *population* and apply the bootstrap sampling to simulate return series.
- Bootstrap is used to take into account sampling uncertainty. The return series obtained from the $j$-th bootstrap sample of $\hat{\mathbf{D}}^i$ is denoted as $\hat{\mathbf{D}}^{i,j}$.

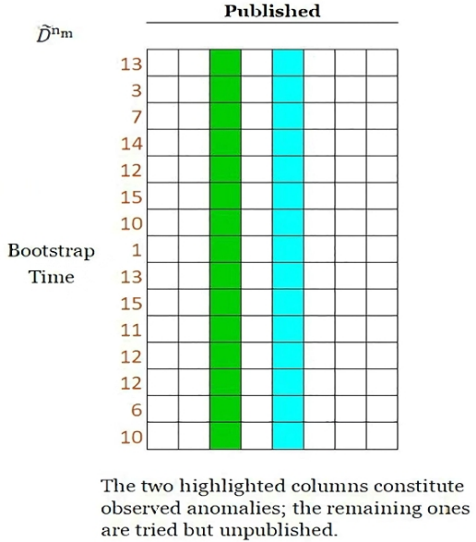



The bootstrapped data will have non-zero means for the initially demeaned data and will alter the means of the untransformed data.



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- For each $\hat{\mathbf{D}}^{i,j}$, calculate the $t$-statistics of all factors and use the formula for prob($t$-statistic) to calculate the publication probability $p_i$.
- For each factor, a uniformly distributed random variable between 0 and 1 is generated; if its value is less than $p_i$, the factor is considered published; otherwise, it is not.
- Let $\tilde{K}_{i,j}$ represent the number of factors ultimately published from bootstrap sample $\hat{\boldsymbol{D}}^{i,j}$, and denote the return series of all published factors as $\tilde{\mathbf{D}}^{i,j}$.




The two highlighted columns constitute observed anomalies; the remaining ones are tried but unpublished.



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- For a given set of parameters $[K_0, p_0, \gamma_1, \gamma_2]$, the method can simulate a set of published factors.
- The choice of parameters is based on how closely certain statistical indicators of the simulated published factors match the actual data. To this end, Harvey and Liu select five quantiles (i.e., 5th, 10th, 50th, 90th, and 95th) of the $t$-statistics of the actual published factors, as well as the number of significant factors, in the calibration.
- The goal is to minimize the following objective function, which is the weighted average of the squared errors between the simulated and actual quantities:

$$\text{obj func} = \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} \left( \sum_{s=1}^{5} \tilde{w}_s \left( \tilde{pc}_s^{i,j} - pc_{0,s} \right)^2 + \tilde{w}_K \left( \tilde{K}_{i,j} - 156 \right)^2 \right),$$

where $\tilde{pc}_s^{i,j}$ and $pc_{0,s}$ represent the $s$-th quantile of $t$-statistic calculated using the bootstrap sample $\tilde{\mathbf{D}}^{n_m}$ and the actual data, respectively.


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# Discussion



- The simulated statistics closely match the actual values under the following three sets of parameters:
  - $K_0 = 1000$, $p_0 = 0.29$, $\gamma_1 = 1.17$, and $\gamma_2 = 0.45$;
  - $K_0 = 2000$, $p_0 = 0.12$, $\gamma_1 = 0.75$, and $\gamma_2 = 0.33$;
  - $K_0 = 5000$, $p_0 = 0.06$, $\gamma_1 = 0.21$, and $\gamma_2 = 0.28$.

  The number of factors studied, $K_0$, equals 1000, 2000, and 5000, respectively.

- Surprising? In fact, the problem itself is **unidentifiable** (i.e., there is a lack of identification). In other words, the optimal parameters are not unique.

- Why? The reason this problem is unidentifiable is precisely because only the published factors are observed, while **the true $K_0$ (the total number of factors attempted) remains unknown.**

- This issue is inevitable when dealing with $p$-hacking. As for the possible value of $K_0$, it depends on the researcher's experience and understanding of the empirical data.

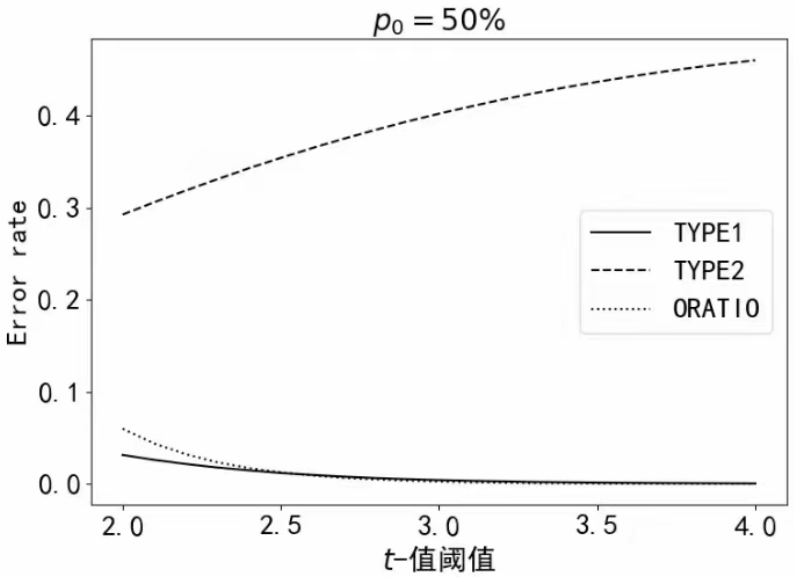

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- There is indirect evidence that tends to support the third set of parameters (i.e., $K_0 = 5000$):
  - The proportion of genuine factors $p_0$ aligns with common sense,
  - The probability of a factor with a $t$-statistic of 2.0 being published aligns with common sense (about 15%).

- With the third set of parameters ($K_0 = 5000$), the model generates the following conclusions:
  1. To control the false discovery rate at 5%, the $t$-statistic cutoff should be 3.0 (this value is consistent with Harvey, Liu and Zhu 2016).
  2. Among all published factors, the proportion of genuine factors is 62%.
  3. Among all published factors, the shrinkage coefficient of the out-of-sample average return is 36%.

- With the simulated publication process, the model can make inferences about the published factors. Under the $K_0 = 5000$ parameter set, 53 false factors get published. They lead to the $p$-hacking problem we see.


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*Questions?*


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